

# Defect-Only Coulomb Truncation for an Isolated Adsorbate on a Periodic Metallic Surface

Method note for O/Ag(111) surface-defect calculations

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## Abstract

A conventional plane-wave calculation of one O adsorbate in an  $N \times N$  Ag(111) surface supercell describes a periodic array of adsorbates. Even with slab Coulomb truncation in the surface-normal direction, the repeated O-induced surface dipoles interact laterally and generate a finite-coverage work-function shift. This note proposes a mixed-boundary electrostatic scheme that keeps pristine Ag(111) periodic in-plane while applying an isolated, Wigner-Seitz-truncated Coulomb interaction only to the defect-induced charge. The method requires reading a converged relaxed no-defect reference density at the start of the defect calculation. At every self-consistent-field (SCF) iteration, the difference between the current defect-system charge and the stored reference charge defines  $\delta\rho$ , from which a correction potential  $(K_{\text{iso}} - K_{\text{slab}})\delta\rho$  is constructed and added to the ordinary Kohn-Sham potential. The resulting electrostatics are intended to approximate a single isolated O adsorbate embedded in an otherwise infinite periodic Ag(111) host.

## 1 Physical problem

The target system is not an ordered oxygen overlayer. It is

$$\text{infinite Ag(111) host} + \text{one isolated O adsorbate.} \quad (1)$$

In an ordinary supercell calculation, however, one O atom in every  $N \times N$  surface cell produces a two-dimensional lattice of defects. For a  $3 \times 3$  cell the nominal coverage is 1/9 monolayer. The O-induced charge transfer creates a surface-normal dipole, and the repeated dipoles produce a macroscopic potential step and a finite-coverage work-function shift. Aligning the vacuum levels of the clean and O-covered calculations therefore leaves the interior Ag potentials displaced.

A standard slab Coulomb cutoff removes interactions between periodic replicas along the surface-normal direction, but it deliberately retains periodicity in the two in-plane directions. Consequently, it cannot remove the lateral electrostatic interaction between periodically repeated adsorbates. A fully isolated Coulomb kernel applied to the *entire* system would remove those interactions, but would also destroy the desired in-plane periodic electrostatics of the Ag substrate.

The proposed solution is therefore to split the electrostatics into a periodic host part and a localized defect-induced part and assign a different Coulomb kernel to each.

## 2 Reference calculation and stored no-defect density

The method begins with a separate, converged reference calculation for relaxed pristine Ag(111). The reference calculation must use the same lateral supercell, lattice vectors, vacuum thickness, FFT grid, pseudopotentials, plane-wave cutoffs, spin treatment, smearing convention, and surface-normal Coulomb treatment as the subsequent defect calculation.

The most important implementation requirement is that the converged no-defect reference density is written to disk and then **read at the start of the defect SCF calculation**. It remains fixed throughout the defect SCF cycle.

For an electrostatically complete implementation, the stored reference should represent the total pseudocharge entering the Poisson problem,

$$\rho_{\text{tot}}^0(\mathbf{r}) = \rho_e^0(\mathbf{r}) + \rho_{\text{ion}}^0(\mathbf{r}), \quad (2)$$

where  $\rho_e^0$  is the converged electronic charge density and  $\rho_{\text{ion}}^0$  is the pseudoionic charge density of the clean Ag slab. If the DFT code stores only the electronic density, the ionic part should be reconstructed from the same pseudopotentials and reference ionic coordinates.

For the first implementation, it is useful to keep the defect geometry fixed while testing the electrostatic correction. Structural relaxation can be added after the energy and force derivatives of the correction have been verified.

## 2.1 Choice of reference ionic geometry

Two reference choices are useful and should not be confused:

- **Relaxed pristine reference:** the Ag atoms are at their relaxed clean-surface coordinates. Then  $\delta\rho$  includes both the electronic response to O and any change of ionic pseudocharge caused by O-induced Ag relaxation.
- **Frozen-host reference:** remove O from the defect-relaxed structure but keep the Ag atoms at the defect-relaxed coordinates, and compute/read the corresponding host-only density. This isolates the electronic and ionic contribution associated more directly with adding O at fixed host geometry. It is more expensive but can be useful for separating electrostatic defect physics from elastic relaxation.

The relaxed pristine reference is the simplest starting point and is the default assumed below.

## 3 Defect-induced charge

At SCF iteration  $n$ , let the current total pseudocharge of the O/Ag system be

$$\rho_{\text{tot}}^{(n)}(\mathbf{r}) = \rho_e^{(n)}(\mathbf{r}) + \rho_{\text{ion}}^{\text{O/Ag}}(\mathbf{r}). \quad (3)$$

The defect-induced total charge is then defined by direct subtraction of the stored no-defect reference,

$$\boxed{\delta\rho_{\text{tot}}^{(n)}(\mathbf{r}) = \rho_{\text{tot}}^{(n)}(\mathbf{r}) - \rho_{\text{tot}}^0(\mathbf{r})}. \quad (4)$$

For neutral reference and defect cells,

$$\int_{\Omega} d^3r \, \delta\rho_{\text{tot}}(\mathbf{r}) = 0, \quad (5)$$

up to numerical integration error. This neutrality should be checked explicitly because the isolated kernel acts most cleanly on a neutral localized perturbation.

Equation (4) is the central reason for reading the converged relaxed clean density from disk rather than rebuilding an approximate reference from isolated atomic densities. The subtraction removes the already self-consistent periodic Ag background and leaves only the charge rearrangement associated with the defect and its screening cloud.

## 4 Mixed-boundary Coulomb functional

Let  $K_{\text{slab}}$  denote the ordinary slab Coulomb operator: periodic in the two surface directions and truncated along the surface normal. Let  $K_{\text{iso}}$  denote a three-dimensionally isolated Coulomb operator, preferably truncated on the three-dimensional Wigner–Seitz cell of the supercell. Define

$$\Delta K \equiv K_{\text{iso}} - K_{\text{slab}}. \quad (6)$$

The desired electrostatic potential can be written as

$$V_{\text{es}}^{\text{desired}} = K_{\text{slab}}\rho_{\text{tot}}^0 + K_{\text{iso}}\delta\rho_{\text{tot}} \quad (7)$$

$$= K_{\text{slab}}\rho_{\text{tot}} + (K_{\text{iso}} - K_{\text{slab}})\delta\rho_{\text{tot}}. \quad (8)$$

Thus an existing slab calculation can be retained almost unchanged. One computes the ordinary slab electrostatic potential and adds only the defect correction

$$\boxed{V_{\text{corr}}(\mathbf{r}) = \Delta K \delta\rho_{\text{tot}}(\mathbf{r}) = (K_{\text{iso}} - K_{\text{slab}})\delta\rho_{\text{tot}}(\mathbf{r})}. \quad (9)$$

The corresponding correction to the electrostatic energy is

$$\boxed{E_{\text{corr}} = \frac{1}{2} \langle \delta\rho_{\text{tot}} | \Delta K | \delta\rho_{\text{tot}} \rangle}. \quad (10)$$

Because the stored reference density is fixed, functional differentiation gives

$$\frac{\delta E_{\text{corr}}}{\delta \rho_{\text{tot}}(\mathbf{r})} = V_{\text{corr}}(\mathbf{r}), \quad (11)$$

so Eqs. (9) and (10) are variationally consistent.

The complete Kohn–Sham potential is therefore the usual potential generated by the O/Ag density under slab electrostatics, plus  $V_{\text{corr}}$ . Exchange–correlation terms, kinetic energy, local short-range pseudopotential terms, and nonlocal pseudopotential projectors continue to use the full physical density and are not Coulomb truncated by this construction.

## 5 SCF algorithm

Figure 1 summarizes the proposed workflow.

A practical iteration is:

1. Read the converged relaxed pristine Ag reference electronic density  $\rho_e^0(\mathbf{r})$  from disk. Reconstruct or read the corresponding reference ionic pseudocharge  $\rho_{\text{ion}}^0(\mathbf{r})$ .
2. Initialize the O/Ag electronic density. A particularly good initial condition is the stored Ag density plus an O atomic contribution, or a previously converged periodic O/Ag density.
3. At SCF iteration  $n$ , construct the current total pseudocharge  $\rho_{\text{tot}}^{(n)}$ .
4. Compute  $\delta\rho_{\text{tot}}^{(n)} = \rho_{\text{tot}}^{(n)} - \rho_{\text{tot}}^0$  on the common FFT grid.
5. Check or enforce numerical neutrality of  $\delta\rho_{\text{tot}}$ . A significant nonzero integral indicates inconsistent electron number, ionic pseudocharge normalization, grid conventions, or reference data.
6. Apply the standard slab kernel to the full current total charge, as in the ordinary code path.

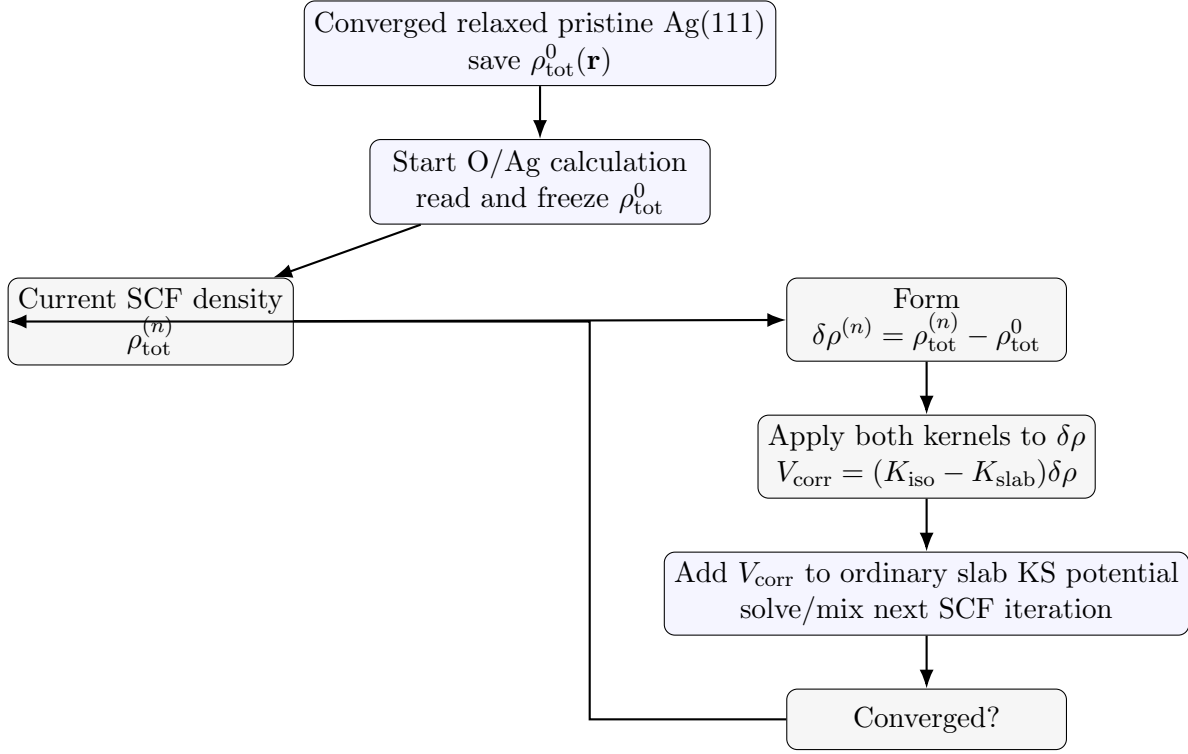


Figure 1: SCF flow for defect-only Coulomb truncation. The relaxed pristine reference density is read once and remains fixed.

7. Apply both the isolated and slab kernels to  $\delta\rho_{\text{tot}}$  and construct

$$V_{\text{corr}}^{(n)} = (K_{\text{iso}} - K_{\text{slab}})\delta\rho_{\text{tot}}^{(n)}.$$

8. Add  $V_{\text{corr}}^{(n)}$  to the electrostatic contribution to the Kohn–Sham potential before solving the next orbital step.
9. Add Eq. (10) to the total energy.
10. Continue ordinary density/potential mixing until the density, total energy, and correction energy are converged.

A compact pseudocode representation is

```

rho_ref_e = read_density("clean_relaxed_density")
rho_ref_ion = build_reference_pseudoionic_density()
rho_ref_tot = rho_ref_e + rho_ref_ion

```

for each SCF iteration:

```

rho_e = current_electronic_density()
rho_ion = current_pseudoionic_density()
rho_tot = rho_e + rho_ion

```

```

delta_rho = rho_tot - rho_ref_tot

```

```

V_slab = K_slab(rho_tot)
V_corr = K_iso(delta_rho) - K_slab(delta_rho)

```

```

V_es          = V_slab + V_corr

E_corr        = 0.5 * inner(delta_rho, V_corr)

build_KS_potential(V_es, XC, local_PP, ...)
solve_orbitals_and_mix_density()

```

## 6 Why the method approximates an isolated O adsorbate

The periodic Ag background is acted on only by  $K_{\text{slab}}$ , so the clean substrate retains the correct two-dimensional periodic boundary condition. By contrast, only the neutral defect-induced charge  $\delta\rho_{\text{tot}}$  is acted on by  $K_{\text{iso}}$ . Therefore the O-induced dipole and screening charge do not electrostatically interact with their lateral periodic images.

In an ordinary finite-coverage calculation, the repeated surface dipoles generate a nonzero dipole moment per unit area and hence a macroscopic work-function shift. In the mixed-kernel calculation, the defect electrostatic potential is instead required to decay away from the localized  $\delta\rho_{\text{tot}}$ . Far from the O atom one therefore seeks

$$V_{\text{O/Ag}}(\mathbf{r}) - V_{\text{Ag}}^0(\mathbf{r}) \longrightarrow 0, \quad (12)$$

within the accuracy allowed by the finite quantum-mechanical supercell.

This is exactly the behavior desired before constructing a localized Wannier defect perturbation

$$\Delta H = H_{\text{O/Ag}} - H_{\text{Ag}}^0. \quad (13)$$

The resulting  $\Delta H$  should decay with lateral distance from O instead of containing a spurious approximately uniform shift inherited from an ordered O overlayer.

## 7 Forces and structural relaxation

For a first implementation, fixed ionic coordinates are strongly recommended. Once the energy and potential correction have been validated, forces can be added consistently.

When ionic pseudocharges are included in  $\delta\rho_{\text{tot}}$ , the correction energy in Eq. (10) depends explicitly on ionic coordinates. The force correction is

$$\mathbf{F}_I^{\text{corr}} = -\frac{\partial E_{\text{corr}}}{\partial \mathbf{R}_I}, \quad (14)$$

including the derivative of the current pseudoionic charge density and, if the reference ionic geometry itself is parameterized, the appropriate reference terms. The stored relaxed pristine reference should normally be treated as fixed; only the current O/Ag ionic pseudocharges vary during a corrected relaxation.

Because the long-range Coulomb correction is the new piece, short-range nonlocal pseudopotential force expressions should remain unchanged. A finite-difference force test on O and on a nearby Ag atom should be required before using the method for geometry optimization.

## 8 Implementation checks

The following diagnostics should be built into the code or post-processing from the beginning:

### 1. Charge neutrality:

$$Q_{\text{defect}} = \int_{\Omega} \delta\rho_{\text{tot}}(\mathbf{r}) d^3r \approx 0. \quad (15)$$

2. **Reference recovery:** if O is removed and the current density equals the stored reference, then  $\delta\rho_{\text{tot}} = 0$ ,  $V_{\text{corr}} = 0$ , and  $E_{\text{corr}} = 0$  to numerical precision.
3. **Potential localization:** after convergence, plot the planar and lateral behavior of

$$\Delta V(\mathbf{r}) = V_{\text{O/Ag}}(\mathbf{r}) - V_{\text{Ag}}^0(\mathbf{r}). \quad (16)$$

It should approach zero in Ag regions far from O and should not exhibit a uniform interior offset caused only by the adsorbate-array work function.

4. **Density localization:** visualize  $\delta\rho_{\text{tot}}(\mathbf{r})$  and verify that the induced charge and screening cloud are well contained inside the Wigner–Seitz truncation region.
5. **Kernel confinement requirement:** the density difference must satisfy the real-space support requirements of the chosen truncated kernel. In JDFTx truncated-interaction modes, the relevant charge separation must remain within half the truncated-cell length in each truncated direction.
6. **Energy consistency:** compare analytic  $V_{\text{corr}}$  against a finite difference of  $E_{\text{corr}}$  with respect to a small trial density perturbation.

## 9 Validation against supercell size

The mixed-kernel method removes electrostatic periodic-image interactions, but it does *not* eliminate all finite-supercell effects. The Kohn–Sham orbitals still obey the supercell translational symmetry, so substrate-mediated O–Ag–O hybridization, impurity-band formation, Friedel oscillations, and elastic interactions can remain.

The method must therefore be validated by comparing at least two or three lateral cells, for example

$$3 \times 3, \quad 4 \times 4, \quad 5 \times 5. \quad (17)$$

The goal is not primarily convergence of the supercell work function. The relevant observable is convergence of the *local* defect Hamiltonian. After Wannierization, compare quantities such as

$$\Delta H_{\text{OO}}, \quad \Delta H_{\text{O,Ag}_1}, \quad \Delta H_{\text{Ag}_1,\text{Ag}_1}, \quad (18)$$

and the Frobenius norm of local onsite blocks as a function of distance from O,

$$D_i = \|\Delta H_{ii}\|_F. \quad (19)$$

Success is indicated when the mixed-kernel  $3 \times 3$  or  $4 \times 4$  calculation gives essentially the same near-defect  $\Delta H$  as a substantially larger conventional or corrected cell, while  $D_i$  decays toward zero before the supercell boundary.

## 10 Suggested implementation sequence in JDFTx

JDFTx already contains slab and isolated Coulomb truncation machinery, so the new development can be kept conceptually small. A staged implementation is recommended:

1. Add an input option specifying a saved relaxed no-defect reference density file.
2. Read the reference electronic density on startup and verify that its grid and cell exactly match the active calculation.
3. Construct the reference ionic pseudocharge from the pristine Ag species and stored reference coordinates.

4. Expose access to both a CoulombSlab and a three-dimensional isolated/Wigner–Seitz Coulomb operator within the same SCF calculation.
5. At each iteration, form  $\delta\rho_{\text{tot}}$  and compute  $V_{\text{corr}}$ .
6. Add  $V_{\text{corr}}$  to the electrostatic Kohn–Sham potential and  $E_{\text{corr}}$  to the total energy.
7. Validate with fixed ions and finite-difference density tests.
8. Add ionic force derivatives and validate with finite differences.
9. Finally, compare corrected  $3 \times 3$ ,  $4 \times 4$ , and  $5 \times 5$  O/Ag(111) calculations and quantify convergence of the local Wannier defect Hamiltonian.

The core modification can be summarized as

$$V_{\text{es}} = K_{\text{slab}}\rho_{\text{tot}} + (K_{\text{iso}} - K_{\text{slab}})(\rho_{\text{tot}} - \rho_{\text{tot}}^0). \quad (20)$$

The stored relaxed no-defect density  $\rho_{\text{tot}}^0$  is therefore not merely an initial guess: it defines the periodic host background that is deliberately excluded from the isolated Coulomb correction.

## 11 Limitations and possible extensions

The method addresses the dominant long-range electrostatic artifact of a periodic adsorbate array. It does not by itself remove finite-cell quantum overlap or elastic image interactions. If those become important, larger cells or a more complete Green-function embedding formulation are still required.

A natural future extension is a grand-canonical metallic boundary condition in which the electron chemical potential is pinned to the pristine Ag reservoir. This would suppress finite-concentration shifts of the metal Fermi level in addition to the electrostatic work-function artifact. Another extension is to combine the mixed-kernel SCF with a Wannier embedding step so that only a compact converged  $\Delta H$  around the adsorbate is retained and embedded in the pristine semi-infinite Ag(111) Green function.

## References

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- [3] JDFTx documentation, “coulomb-interaction”: slab, cylindrical, wire, isolated, and spherical truncation modes, [jdftx.org/CommandCoulombInteraction.html](http://jdftx.org/CommandCoulombInteraction.html).